Statistical Learning Exercices 1

Exercice 1. *Assume that we have a set of data that describes the following three variables:* $Hair = \{B, D\}$, *where* $B = blond, D = dark$. $Height = \{T, S\}$, *where* $T = tall, S = short$. $Country = \{G, P\}$, *where* $G = Greenland, P = Poland$. *You are given the following workout data set (Hair, Height, Country):*

$$\begin{pmatrix} (B,T,G) & (B,T,G) & (B,T,P) \\ (D,T,G) & (D,T,G) & (B,T,P) \\ (D,T,G) & (D,T,G) & (B,T,P) \\ (D,T,G) & (D,T,G) & (D,T,P) \\ (B,T,G) & (B,T,G) & (D,T,P) \\ (B,S,G) & (B,S,G) & (D,S,P) \\ (B,S,G) & (B,S,G) & (B,S,P) \\ (D,S,G) & (D,S,G) & (D,S,P) \end{pmatrix}$$

We want to answer the following question: if you observe a new tall individual having blond hair, what is her most likely country of origin?

1. *Give the maximum posterior solution (MAP) to the question, using the naive frame from Bayes. Detail your work.*
2. *Give the maximum a posteriori solution (MAP) to the question, without using the naive frame from Bayes. Detail your work.*
3. *Give the maximum likelihood solution (MLE) to the question, using the naive frame from Bayes. Detail your work.*
4. *Give the maximum likelihood solution (MLE) to the question, without using the naive frame from Bayes. Detail your work.*

Exercice 2. *We consider a tiny data set with* 10 *observations* $(x_i, y_i)_{1 \leq i \leq 10}$, *with* $x_i \in \mathcal{X}$ *and* $y_i \in \{-1, 1\}$. *Using different machine learning algorithms, two models are built from this data set:* g_1 *and* g_2. *Recall that a model is a function from* $\mathcal{X}$ *to* $\{-1, 1\}$. *The outputs of the models on the learning data set are given by the following table:*

x_i	$g_1(x_i)$	$g_2(x_i)$	y_i
x_1	1	1	1
x_2	1	-1	1
x_3	-1	1	1
x_4	1	-1	1
x_5	1	-1	1
x_6	1	-1	-1
x_7	-1	-1	-1
x_8	-1	-1	-1
x_9	-1	1	-1
x_{10}	1	-1	-1

1. *Compute the confusion matrices of both models on the learning set.*

2. *We use the loss function ℓ_1 given by:*

$\ell_1(p,t)$	*t=-1*	*t=1*
p=-1	*0*	*1*
p=1	*3*	*0*

where p is the predicted value and t the true value. Compute the empirical risk of both models on the learning set for ℓ_1.

3. *Determine the best model based on the available information using the loss function $L_b(p,t) = \mathbf{1}_{p \neq t}$.*

Exercice 1. *Assume that we have a set of data that describes the following three variables: Hair = $\{B, D\}$, where B = blond, D = dark. Height = $\{T, S\}$, where T = tall, S = short. Country = $\{G, P\}$, where G = Greenland, P = Poland. You are given the following workout data set (Hair, Height, Country):*

$$\begin{pmatrix} (B,T,G) & (B,T,G) & (B,T,P) \\ (D,T,G) & (D,T,G) & (B,T,P) \\ (D,T,G) & (D,T,G) & (B,T,P) \\ (D,T,G) & (D,T,G) & (D,T,P) \\ (B,T,G) & (B,T,G) & (D,T,P) \\ (B,S,G) & (B,S,G) & (D,S,P) \\ (B,S,G) & (B,S,G) & (B,S,P) \\ (D,S,G) & (D,S,G) & (D,S,P) \end{pmatrix}$$

We want to answer the following question: if you observe a new tall individual having blond hair, what is her most likely country of origin?

1. *Give the maximum posterior solution (MAP) to the question, using the naive frame from Bayes. Detail your work.*

L'idée est de comparer $\mathbb{P}(G \mid B,T)$ et $\mathbb{P}(P \mid B,T)$

Or puisqu'on est dans le cadre naïf de Bayes,

Greenland Bayes

① $$\mathbb{P}(G \mid B,T) = \frac{\mathbb{P}(B,T \mid G) \times \mathbb{P}(G)}{\mathbb{P}(B,T)}$$

naïf ② $$\mathbb{P}(B,T) = \mathbb{P}(B) \times \mathbb{P}(T) \quad \& \quad \mathbb{P}(B,T \mid G) = \mathbb{P}(B \mid G) \times \mathbb{P}(T \mid G)$$

$$\Rightarrow \mathbb{P}(G \mid B,T) = \frac{\mathbb{P}(B \mid G) \times \mathbb{P}(T \mid G) \times \mathbb{P}(G)}{\mathbb{P}(B) \times \mathbb{P}(T)} \leftarrow$$

(Poland $$\mathbb{P}(P \mid B,T) = \frac{\mathbb{P}(B \mid P) \times \mathbb{P}(T \mid P) \times \mathbb{P}(P)}{\mathbb{P}(B) \times \mathbb{P}(T)} \leftarrow$$)

dénom idem donc on compare que numérateurs :

Or on a $\mathbb{P}(B \mid G) = 8/16 = 1/2$.
$\mathbb{P}(T \mid G) = 10/16 = 5/8$.
$\mathbb{P}(G) = 16/24 = 2/3$.

$$\Rightarrow \mathbb{P}(G \mid B,T) = \frac{\mathbb{P}(B \mid G) \times \mathbb{P}(T \mid G) \times \mathbb{P}(G)}{\mathbb{P}(B) \times \mathbb{P}(T)}$$

$$= \frac{\frac{1}{2} \times \frac{5}{8} \times \frac{2}{3}}{\mathbb{P}(B) \times \mathbb{P}(T)} = \frac{10/48 = 0{,}2083}{\mathbb{P}(B) \times \mathbb{P}(T)}$$

$4/8 = 1/2 \times 5/8 \times 1/3 = 5/48 = 0{,}1042$

$$\mathbb{P}(P \mid B,T) = \frac{\mathbb{P}(B \mid P) \times \mathbb{P}(T \mid P) \times \mathbb{P}(P)}{\mathbb{P}(B) \times \mathbb{P}(T)}$$

donc $\mathbb{P}(G \mid B,T) > \mathbb{P}(P \mid B,T)$

donc la sol° du max a posteriori dans le cadre naïf de Bayes est Greenland

2. *Give the maximum a posteriori solution (MAP) to the question, without using the naive frame from Bayes. Detail your work.*

MAP sans le cadre Naïf de Bayes :

$$\mathbb{P}(G \mid B,T) = \frac{\mathbb{P}(B,T \mid G) \times \mathbb{P}(G)}{\mathbb{P}(B,T)} \quad \text{et} \quad \mathbb{P}(P \mid B,T) = \frac{\mathbb{P}(B,T \mid P) \times \mathbb{P}(P)}{\mathbb{P}(B,T)}$$

(ce qui revient à comparer $\mathbb{P}(B,T \mid G) \times \mathbb{P}(G)$ et $\mathbb{P}(B,T \mid P) \times \mathbb{P}(P)$:

Or $\mathbb{P}(B,T \mid G) = 4/16 = 1/4$. $\mathbb{P}(G) = 2/3$ $\Big\} \mathbb{P}(B,T \mid G) \times \mathbb{P}(G) = 1/6 = 0,1667$

et $\mathbb{P}(B,T \mid P) = 3/8$ $\mathbb{P}(P) = 1/3$ $\Big\} \mathbb{P}(B,T \mid P) \times \mathbb{P}(P) = \frac{3}{24} = 0,125$

Donc $\mathbb{P}(B,T \mid G) \times \mathbb{P}(G) > \mathbb{P}(B,T \mid P) \times \mathbb{P}(P)$

Donc $\mathbb{P}(G \mid B,T) > \mathbb{P}(P \mid B,T)$

Donc la sd° du MAP sans le cadre naïf de Bayes est Greenland

3. *Give the maximum likelihood solution (MLE) to the question, using the naive frame from Bayes. Detail your work.*

MLE dans le cadre naïf de Bayes :

↳ on compare seulement les termes de vraisemblance :

$$\mathbb{P}(B,T \mid G) \quad \text{et} \quad \mathbb{P}(B,T \mid P)$$

Or $\mathbb{P}(B,T \mid G) \overset{\text{naïf}}{=} \mathbb{P}(B \mid G) \times \mathbb{P}(T \mid G) = \frac{1}{2} \times \frac{5}{8} = \frac{5}{16}$

$\mathbb{P}(B,T \mid P) \overset{\text{naïf}}{=} \mathbb{P}(B \mid P) \times \mathbb{P}(T \mid P) = \frac{1}{2} \times \frac{5}{8} = \frac{5}{16}$

Donc on ne fait pas prendre de décision.

4. *Give the maximum likelihood solution (MLE) to the question, without using the naive frame from Bayes. Detail your work.*

On compare juste $\mathbb{P}(B,T \mid G)$ et $\mathbb{P}(B,T \mid P)$
$= 1/4 \quad < \quad = 3/8$

Donc d'après le MLE sans le cadre naïf de Bayes, sd° = Poland

Exercice 2. *We consider a tiny data set with 10 observations* $(x_i, y_i)_{1 \leq i \leq 10}$, *with* $x_i \in \mathcal{X}$ *and* $y_i \in \{-1, 1\}$. *Using different machine learning algorithms, two models are built from this data set:* g_1 *and* g_2. *Recall that a model is a function from* $\mathcal{X}$ *to* $\{-1, 1\}$. *The outputs of the models on the learning data set are given by the following table:*

x_i	$g_1(x_i)$	$g_2(x_i)$	y_i
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x_3	−1	1	1
x_4	1	−1	1
x_5	1	−1	1
x_6	1	−1	−1
x_7	−1	−1	−1
x_8	−1	−1	−1
x_9	−1	1	−1
x_{10}	1	−1	−1

1. *Compute the confusion matrices of both models on the learning set.*

matrices de confus° :

PREDICTION \ REALITÉ (model g_2)	neg −1	pos 1
neg −1	TRUE NEG 3	FAUX NEG 1
pos 1	FAUX POS 2	TRUE POS 4

PREDICTION \ REALITÉ (model g_2)	neg −1	pos 1
neg −1	TRUE NEG 4	FAUX NEG 3
pos 1	FAUX POS 1	TRUE POS 2

2. *We use the loss function* ℓ_1 *given by:*

$\ell_1(p,t)$	t=-1	t=1
p=-1	0	1
p=1	3	0

where p *is the predicted value and* t *the true value. Compute the empirical risk of both models on the learning set for* ℓ_1.

Risque Empirique : $\hat{R}_L(g, D) = \frac{1}{N} \sum_{i=1}^{N} L(g(X_i), Y_i)$

coût

↳ on chq obs° i on compare $g(x_i)$ à la vraie val Y_i via f° de coût L puis moyenne sur les N obs.

énoncé

↳ $Y = \{-1, 1\}$ (2 et 3) $\quad N = 10$

↳ Ĉ il n'y a que 2 C ftes, il n'y a que 4 combinaisons pour (prédict°, réalité) : (-1,-1) (1,-1) (-1,1) (1,1)

↳ grâce à matrice de confus°, on pt regrp par catég :

$$R_{g_1^{g_2}} = \frac{1}{N}\sum_{i=1}^{N} L(g_2(x_i), y_i) = \frac{1}{N}\sum_{p\in Y}\sum_{t\in Y} g_2^{g_1}(p,t)\, l_1(p,t)$$

$$R_{g_1} = \frac{1}{10}(3\times 0 + 1\times 1 + 2\times 3 + 4\times 0) = \frac{7}{10} = 0{,}7$$

$$R_{g_2} = \frac{1}{10}(4\times 0 + 3\times 1 + 1\times 3 + 2\times 0) = \frac{6}{10} = 0{,}6$$

3. *Determine the best model based on the available information using the loss function* $L_b(p,t) = \mathbf{1}_{p\neq t}$.

$L_b(p,t) = \mathbf{1}_{p\neq t}$ revient à :

$L_b(p,t)$	$t=-1$	$t=1$
$p=-1$	0	1
$p=1$	1	0

$$R_{g_2^{g_1}} = \frac{1}{N}\sum_{p\in Y}\sum_{t\in Y} g_2^{g_1}(p,t)\, L_b(p,t)$$

$R_{g_1} = \frac{3}{10}$ $\quad$ $R_{g_2} = \frac{4}{10}$